

Content Delivery Performance Analysis of a Cache-Enabled UAV Base Station Assisted Cellular Network for Metaverse Users

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Abstract—Metaverse can provide powerful human-centric interactive experiences for users and metaverse application content is the fundamental component that supports the metaverse. Considering that metaverse users are more sensitive to the delay of content delivery service, unmanned aerial vehicles (UAV) can be used as aerial base stations (BSs) to assist a cellular network to provide better content delivery service for delay-sensitive metaverse users as UAV base stations (UBSs) have big potential for line-of-sight (LoS) transmission and can be deployed closer to metaverse users than macro base stations (MBSs). This paper studies the content delivery performance analysis of a cache-enabled UBS-assisted cellular network for metaverse users. Analytical models are derived for investigating the content delivery performance of the network in terms of the content delivery success probability of the network and the average content delivery delay of a metaverse user. In deriving the analytical models, a more realistic repulsive point process is considered for modeling the location distribution of MBSs, and an air-to-ground (A2G) channel model with both a LoS link and a non-line-of-sight (NLoS) link is considered. Moreover, the cache hit probability of a UBS using a probabilistic caching strategy is also taken into consideration. A BS association strategy for delay-sensitive metaverse users based on the strongest average received power at a user and the cache hit probability of a UBS is proposed. In addition, the association probabilities with the association strategy are derived for different types of base stations. A lower bound of the content delivery success probability and an upper bound of the average content delivery delay are obtained based on the derived analytical models. The numerical results justify the effectiveness and advantage of the proposed BS association strategy and show that there exist an optimal UBS height and an optimal value of the number of UBSs, which result in the optimal content delivery performance. The obtained results can provide theoretical guidance for the deployment of UBSs in a cache-enabled UBS-assisted cellular network to provide better human-centric content delivery service for metaverse user.

Index Terms—UAV base station, content delivery, performance analysis, cellular network, metaverse user.

I. INTRODUCTION

METaverse is a massive virtual environment parallel to the physical world [1], in which humans are usually the main users of most metaverse applications. Thus, human-centric is one of the unique features of the metaverse. In the metaverse, users can explore the virtual world and interact with the physical world through the aid of augmented reality (AR), virtual reality (VR), and the tactile Internet [2]. However, such powerful human-centric interactive experiences require strong support from existing computing and communication infrastructure, especially in high-revolution content delivery, and ultra-low and uniform network latency, which exceeds the capability of existing network infrastructure [3]. Considering that metaverse application content is the fundamental component that supports the metaverse, it is necessary to enhance the network performance to provide better content delivery service for users in the metaverse.

Unmanned Aerial Vehicles (UAVs) can be used as aerial base stations (BSs) to assist a cellular network to provide content delivery service for metaverse users [4]. UAV base stations (UBSs) have a big potential for line-of-sight (LoS) transmission [5], which can improve channel transmission quality and reduce content delivery delay. Moreover, UBSs can carry cached content closer to the users than cellular ground BSs due to their flexibility. Therefore, a cache-enabled UBS-assisted cellular network can provide users with better human-centric content delivery service in the metaverse [6]. In particular, users in the metaverse usually need real-time interactions. They are more sensitive to the delay of content delivery service compared with normal users. Therefore, human-centric content delivery service should be provided with stringent delay requirements, and it is important to analyze and improve the content delivery performance for delay-sensitive metaverse users in a cache-enabled UBS-assisted cellular network.

This paper studies the content delivery performance analysis of a cache-enabled UBS-assisted cellular network for metaverse users. Analytical models are derived for investigating the content delivery performance of the network in terms of the content delivery success probability of the network and the average content delivery delay of a delay-sensitive user. In deriving the analytical models, a more realistic repulsive point

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process is considered for modeling the location distribution of macro base stations (MBSs), and an air-to-ground (A2G) channel model with both a LoS link and a non-line-of-sight (NLoS) link is considered. Moreover, the cache hit probability of a UBS using a probabilistic caching strategy is also taken into consideration. Considering that metaverse users are more sensitive to the delay of content delivery service compared with normal users, we propose a BS association strategy for delay-sensitive users based on the strongest average received power at a user and the cache hit probability of a UBS. In addition, the association probabilities with the proposed BS association strategy are derived for different types of base stations. Finally, a lower bound of the content delivery success probability and an upper bound of the average content delivery delay are obtained based on the derived analytical models. The derived analytical models are validated, and the effectiveness and advantage of the proposed BS association strategy is justified through simulation results and the impacts of system parameters on the content delivery performance are investigated through numerical results.

The next section reviews related work. Section III describes the considered system model. Section IV presents the derivation of the analytical performance models. Section V validates the derived analytical models through simulation results and presents performance investigation through numerical results. Section VI concludes this paper.

II. RELATED WORK

An immersive, embodied, and interoperable metaverse requires support from existing 5G network infrastructure and beyond. Real-time AR/VR content delivery is needed for metaverse users to share a 3D virtual world and AR/VR content can be deployed in a mobile edge network [2]. To provide better content delivery services for metaverse users, a variety of relevant issues have been studied in the literature. For example, Zhang et al. studied location-dependent AR services in a wireless multi-access edge computing-enabled metaverse system and proposed a joint resolution and transmission power optimization algorithm for a high quality of experience (QoE) in the downlink phase in [7]. In [8], Hou et al. designed several bandwidth reservation schemes according to the traffic status of mobile users and formulated an optimization problem for minimizing the reserved bandwidth under service and reliability constraints. Based on unsupervised learning, they further proposed a data-driven method to improve the longitudinal spectrum efficiency in a wireless network. In [9], Gao et al. studied dynamic probabilistic caching based on time-varying instantaneous content popularity. Specifically, they designed probabilistic content placement and replacement policies with the objective to increase the cache hit ratio, and established a novel connection between a caching probability and a probabilistic content placement policy through a mixed strategy. In [10], Li et al. formulated a joint optimization problem for the design of a virtual network (VN) topology and VN embedding (VNE) with the objective to minimize the VNE cost. A couple of algorithms are proposed to solve the formulated problem for substrate networks of small and large scales, respectively. In [11], Wang et al. proposed an intelligent

network-slicing architecture based on digital twin (DT), and proposed a DT based on an inductive graph framework to generate feature embeddings of the network slices represented as graphs, which are used to capture the interconnection relationships between the network slices in different physical network environments. In [12], Sun et al. presented a wireless VR service system that incorporates an intelligent controller, an open 5G BS, and a fog VR server implemented with Air Light VR (ALVR). The results show theoretically that fog computing results in a lower latency as compared to traditional cloud computing. However, none of the above work considered the use of a UAV as a UBS to provide better content delivery service for metaverse users than cellular ground BSs.

In the context of content delivery service, content delivery performance analysis of a UAV-assisted cellular network has been widely studied in the literatures [13], [14], [15], [16], [17], and [18]. In [13], Khuwaja et al. developed a hybrid caching network comprising UAVs and ground small-cell base stations. The successful content delivery probability of the network is derived, and the energy efficiency of the network is analyzed. Moreover, a caching strategy is further proposed to improve the successful content delivery probability. In [14], Lin et al. proposed a caching placement strategy based on the probabilistic caching placement in a heterogeneous wireless network. The derived average service success probability models are based on the cache hit probability and the successful transmission probability. The average service success probability is maximized by optimizing the probability caching placement with a genetic algorithm. In [15], Wang et al. investigated UAV-assisted cellular networks by considering the effect of both the mmWave backhaul and edge caching, and also analyzed the successful content delivery performance and evaluated the impact of key parameters, such as the altitude and density of UAVs, on the successful content delivery performance. The focus of the above work is on the performance modeling and analysis of the network.

Another focus of existing relevant work is on the performance optimization of the network. In [16], Zhou et al. proposed a low-latency VR delivery system where a UBS is deployed to deliver VR content from a cloud server to multiple ground VR users. Moreover, they designed a low-complexity iterative algorithm to minimize the maximum communication and computing latency. The numerical results indicate that the proposed algorithm can achieve a lower latency compared to other benchmark algorithms. In [17], Li et al. proposed a distributed delay optimization algorithm which uses massive UAVs to cache the request contents of users. Moreover, they used the robust mean field game theory to model a caching and dynamic flight strategy problem. The simulation results show that the proposed algorithm can reduce the download delay under limited energy consumption. In [18], Zhang et al. studied the joint optimization problem of UAV deployment, caching placement, and user association in a UAV-assisted cellular network, and proposed a low-complexity suboptimal algorithm to solve the problem. However, none of the above work analyzed the content delivery performance of a delay-sensitive user in a cache-enabled UBS-assisted cellular network.

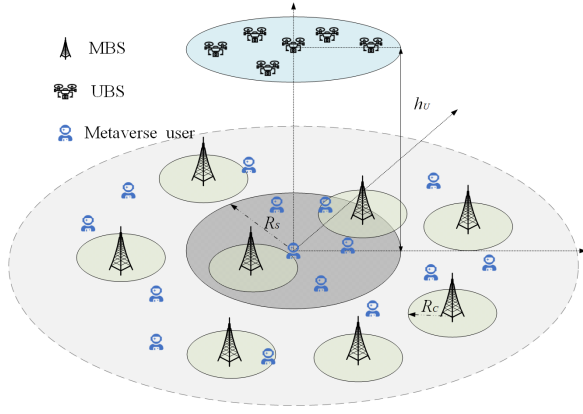


Fig. 1. Network scenario.

III. SYSTEM MODEL

A. Network Scenario

Consider a cache-enabled UBS-assisted cellular network, which consists of several MBSs, N_U UBSs, and a number of ground metaverse users, as shown in Fig. 1. The ground metaverse users are randomly distributed and their location distribution follows a Poisson point process (PPP) with density λ_u . The MBSs and UBSs provide content delivery service for the metaverse users. Compared with a normal user, a ground metaverse user is more sensitive to the delay of content delivery service. A metaverse user can access either an MBS or a UBS for service according to an association strategy. Considering that MBSs are usually deployed at a distance, we assume the location distribution of the MBSs follows a repulsive point process, a Matérn hard-core point process (MHCPP) of type II, where the distance between any two MBSs is larger than a certain value R_C . According to [19], the MHCPP of type II is generated by a dependent thinning of a stationary PPP with density λ_P , where each point in the PPP is identified by an independent random number. Specifically, a thinning operation of the PPP is implemented by only retaining the point which has the smallest number among all points in a circle centered at this point with radius R_C . The location distribution of the MBSs is denoted by $\Phi_M = \{M_i\}$, where M_i denotes both an MBS and its location. Assume that each MBS is equipped with an omnidirectional antenna and the transmission power of the omnidirectional antenna is P_M .

The UBSs are deployed in a finite disk with radius at height h_U . Their location distribution follows a Binomial point process (BPP) and is denoted by $\Phi = \{U_j, j=1: N_U\}$, where U_j denotes both a UBS and its location. Moreover, each UBS is equipped with an omnidirectional antenna. The transmission power of the omnidirectional antenna is P_U . In addition, the MBSs and UBSs share the spectrum resources on the access links between BSs and metaverse users.

B. Channel Model

For the transmission link between a metaverse user and a UBS, it can be either LoS or NLoS. Thus, we consider an A2G channel model proposed in [20], in which both a LoS

link and an NLoS link are considered. A UBS which provides a LoS link is called a LoS BS (LBS). A UBS which provides an NLoS link is called an NLoS BS (NBS).

We look at a reference metaverse user located at the center of the projection of the disk where the UBSs are deployed, which is denoted by O and set as the origin of a 3-D coordinate system. According to [21], the probability of the LoS link between a UBS and the reference user is given by

$$P_L(d_{rU}) = \frac{1}{1 + a \exp\left(-b \left[\frac{180}{\pi} \sin^{-1}\left(\frac{h_U}{d_{rU}}\right) - a\right]\right)}, \quad (1)$$

where a and b are environment-dependent constants, d_{rU} denotes the distance between the reference user and the UBS. Thus, the probability of the NLoS link between the UBS and the reference user can be expressed as

$$P_N(d_{rU}) = 1 - P_L(d_{rU}). \quad (2)$$

The A2G channel under consideration is characterized by both path loss and fast fading. Thus, the signal power received from an LBS and that received from an NBS can, respectively, be expressed as

$$P_R^L = P_U d_{rU}^{-\alpha_L} g_L, \quad (3)$$

$$P_R^N = P_U d_{rU}^{-\alpha_N} g_N, \quad (4)$$

where α_L and α_N denote the pass-loss exponents of a LoS link and an NLoS link, respectively; g_L and g_N denote the fast fading power gains of a LoS link and an NLoS link, respectively. According to [20], we assume that the fast fading of a LoS link is characterized by Nakagami- m fading. The fast fading of an NLoS link is characterized by Rayleigh fading. Thus, g_L follows a gamma distribution with a shape parameter m and a scale parameter $1/m$. g_N follows an exponential distribution with unit mean. Accordingly, the complementary cumulative distribution function (CCDF) of g_L and that of g_N can, respectively, be expressed as

$$\bar{F}_{g_L}(x) = P(g_L > x) = \sum_{k=0}^{m-1} \frac{(mx)^k}{k!} \exp(-mx), \quad (5)$$

$$\bar{F}_{g_N}(x) = P(g_N > x) = \exp(-x). \quad (6)$$

For the transmission link between the reference user and an MBS, we consider a traditional ground-to-ground (G2G) channel model [20] and thus the signal power received from the MBS at the reference user is given by

$$P_R^M = P_M d_{rM}^{-\alpha_M} g_M, \quad (7)$$

where d_{rM} denotes the Euclidean distance between the reference user and the MBS, and g_M follows an exponential distribution with unit mean. The CCDF of g_M is expressed as $\bar{F}_{g_M}(x) = P(g_M > x) = \exp(-x)$.

C. Cache Model

Assume that an MBS is connected to a core network through an optical fiber. Thus, an MBS can provide any content requested by a metaverse user. For a cache-enabled UBS, it can directly provide content delivery service to a metaverse user when the content requested by the user is pre-cached in the UBS. Otherwise, the UBS needs to backhaul to a core network to obtain the requested content. Assume that the content library in a core network contains C_{MAX} files, which are denoted by $F = \{f_1, f_2, \dots, f_{C_{MAX}}\}$, and the popularity of the C_{MAX} files follows a Zipf distribution [22]. Here, the popularity of a file is measured by the probability that the file is requested by all users in the network. Thus, the popularity or the requested probability of a file is given by

$$\rho_c = c^{-\alpha} / \sum_{c=1}^{C_{MAX}} c^{-\alpha}, \quad (8)$$

where α denotes the skewness factor of the Zipf distribution, and c denotes the index of the file. According to Eq. (8), the requested probability of a file does not depend on the size of the file. Thus, we assume that all files have the same size, which is normalized to 1. Moreover, we assume that all the UBSs use the same probabilistic caching strategy proposed in [23] and would not update their pre-cached content when they provide services. Let C_U denote the maximum number of files that a UBS can cache, which is called the cache capacity of a UBS. Let $\mathbf{q} = [q_1, q_2, \dots, q_{C_{MAX}}]$ denote the probabilistic caching strategy used, where q_c denotes the probability that file f_c is cached by all the UBSs. According to the caching strategy, there are C_0 most-popular content (MPC) files and $C_U - C_0$ less-popular content (LPC) files cached in all the UBSs, where the cache probability of the MPC files is set to 1, i.e., $q_c = 1, \forall 1 \leq c \leq C_0$ and the cache probability of the LPC files is set to $q_c = \min\left(\frac{\rho_c(C_U - C_0)}{1 - \sum_{k=1}^{C_0} \rho_k}, 1\right), \forall C_0 + 1 \leq c \leq C_{MAX}$. Thus, the caching strategy should meet the following condition:

$$\sum_{c=1}^{C_{MAX}} q_c \leq C_U. \quad (9)$$

For a UBS that a reference user is associated with, define the cache hit probability of the UBS as the probability that the UBS caches the files requested by the user and use P_{hit} to denote the probability. Thus, we have

$$P_{hit} = \sum_{c=1}^{C_{MAX}} \rho_c q_c. \quad (10)$$

D. BS Association Strategy

A metaverse user is usually associated with a BS according to an association strategy when the user requests for a specific content. If a UBS does not have the requested content, the UBS needs to obtain the requested content from a core network, which is called the cache miss case. In the cache miss case, if a reference user is connected to a UBS, the requested content needs to be transmitted from an MBS to

the UBS first and then be transmitted from the UBS to the reference user, which will cause a relatively large delay. Thus, considering that a metaverse user is more sensitive to the delay of content delivery, we introduce a BS association strategy which associates a metaverse user only with the nearest MBS, not with a UBS, for the cache miss case.

If a UBS has the requested content, the UBS can provide the requested content for the reference user directly, which is called the cache hit case. In the cache hit case, both the MBSs and UBSs can provide the specific content that is requested by the reference user. Since there are three types of BSs, i.e., MBS, LBS, and NBS, and the reference user can be associated with any of them, the signal power received from the nearest BS could be smaller than that received from a further BS due to different channel conditions with different types of BSs. Thus, we introduce a BS association strategy which associates a metaverse user with the BS that provides the strongest average received power [24] for the cache hit case.

In summary, the proposed BS association strategy includes the following main steps: 1) the reference user requests for a specific content from BSs and then estimates whether a UBS can provide the requested content directly; 2) if a UBS cannot provide the requested content directly, the reference user will be associated with the nearest MBS; 3) if a UBS can provide the requested content directly, the reference user will be associated with the MBS or UBS which provides the strongest average received power.

IV. PERFORMANCE ANALYSIS

In this section, we consider the downlink content delivery performance of the network with the proposed BS association strategy. Specifically, we derive analytical models for the content delivery success probability of the network and the average content delivery delay of a reference metaverse user, respectively. Moreover, a lower bound of the content delivery success probability and an upper bound of the average content delivery delay are derived in this section. According to the Slivnyak-Mecke theorem [25], the downlink performance of a reference metaverse user can represent the statistical performance of the overall network.

A. Distance Distribution

We first derive the distribution of the distance between a reference user and a UBS, and that of the distance between a reference user and the nearest MBS, respectively. Let D_{rU} denote the distance between the reference user and a UBS. According to [26], the probability density function (PDF) of D_{rU} is given by

$$f_{D_{rU}}(d_{rU}) = \frac{2d_{rU}}{R_s^2}, \quad h_U \leq d_{rU} \leq x_h, \quad (11)$$

where $x_h = \sqrt{R_s^2 + h_U^2}$.

Let D_{rM} denote the distance between the reference user and the nearest MBS. According to [19], the cumulative distribution function (CDF) of D_{rM} can be derived from the

empty probability of a MHCPP. Thus, the CDF of D_{rM} is given by

$$F_{D_{rM}}(d_{rM}) = P(D_{rM} < d_{rM}) = \exp(-\lambda_M \pi d_{rM}^2), \quad (12)$$

where $\lambda_M = \frac{1 - \exp(-\lambda_P \pi R_C^2)}{\pi R_C^2}$. The PDF of D_{rM} is given by

$$f_{D_{rM}}(d_{rM}) = \frac{dF_{D_{rM}}(d_{rM})}{dd_{rM}} = 2\pi\lambda_M d_{rM} e^{-\pi\lambda_M d_{rM}^2}. \quad (13)$$

B. Association Probability

Association probability is defined as the probability that a reference metaverse user is connected to a BS of a particular type. According to the earlier discussion, there exist three types of BSs in the network, i.e., MBS, LBS, and NBS. Thus, we use M, L, and N to denote the corresponding BS types, respectively, and use A_M , A_L , and A_N to denote the corresponding association probabilities, respectively.

According to the association strategy introduced in Section III-D, the reference user is supposed to be connected to the nearest MBS in the cache miss case. Thus, A_M is given by

$$A_M = (1 - P_{hit}) + P_{hit} A_M^{hit}, \quad (14)$$

where A_M^{hit} denotes the probability that the reference user is connected to an MBS in the cache hit case.

In the cache hit case, the reference user is supposed to be connected to the BS which provides the largest long-term average received signal power. Thus, the effect of fast fading can be ignored here. Let X denote a typical BS, $X \in \{M, L, N\}$. The long-term average signal power received by the reference user from X is given by

$$P_{AVR}^X = P_T^X (d_X)^{-\alpha_X}, \quad (15)$$

where P_T^X denotes the transmission power of the omnidirectional antenna of X and d_X denotes the distance between the reference user and X , α_X denotes the pass-loss exponents of the link between X and the reference user.

Let E_0 denote the event that the average signal power received from the MBS is larger than that from a particular UBS. Thus, the probability that event E_0 occurs can be expressed as

$$\begin{aligned} P(E_0) &= \mathbb{P}(P_{AVR}^M > P_{AVR}^L) + \mathbb{P}(P_{AVR}^M > P_{AVR}^N) \\ &= \mathbb{P}(d_L > D_{ML}(d_{rM})) + \mathbb{P}(d_N > D_{MN}(d_{rM})) \\ &= \int_{D_{ML}(d_{rM})}^{x_h} P_L(x) f_{D_{rU}}(x) dx \\ &\quad + \int_{D_{MN}(d_{rM})}^{x_h} P_N(x) f_{D_{rU}}(x) dx, \end{aligned} \quad (16)$$

where $D_{XY}(d_X) = d_X^{\frac{\alpha_X}{\alpha_Y}} (P_T^Y / P_T^X)^{\frac{1}{\alpha_Y}}$ denotes the lower bound of the distance between Y and the reference user under the condition that the average power received by the reference user from X is larger than that from Y , and Y denotes another BS, $X, Y \in \{M, L, N\}$. Considering that the locations of

N_U UBSs are independent and identically distributed (i.i.d.), we have

$$\begin{aligned} A_M^{hit} &= \int_0^{D_{MAX}^M} f_{D_{rM}}(d_{rM}) P(E_0)^{N_U} dd_{rM} \\ &= \int_0^{D_{MAX}^M} f_{D_{rM}}(d_{rM}) \left(\int_{D_{ML}(d_{rM})}^{x_h} P_L(x) f_{D_{rU}}(x) dx \right. \\ &\quad \left. + \int_{D_{MN}(d_{rM})}^{x_h} P_N(x) f_{D_{rU}}(x) dx \right)^{N_U} dd_{rM}, \end{aligned} \quad (17)$$

where $D_{MAX}^M = \max(D_{LM}(x_h), D_{NM}(x_h))$.

For the association probability with an LBS in the cache hit case, we first consider the probability that the reference user is associated with a particular LBS located at a distance r to the reference user, which occurs when the following two events occur simultaneously: 1) the average signal power received from this LBS is larger than that received from the nearest MBS; 2) the average signal power received from this LBS is the larger than that received from other UBSs. Let E_1 and E_2 denote event 1) and event 2), respectively. Thus, we have

$$P(E_1) = \mathbb{P}(D_{rM} > D_{LM}(r)) = \exp(-\pi\lambda_M D_{LM}^2(r)), \quad (18)$$

$$\begin{aligned} P(E_2) &= \left[\int_r^{x_h} P_L(x) f_{D_{rU}}(x) dx \right. \\ &\quad \left. + \int_{D_{LN}(r)}^{x_h} P_N(x) f_{D_{rU}}(x) dx \right]^{N_U - 1}. \end{aligned} \quad (19)$$

Considering that the locations of N_U UBSs are i.i.d., we have

$$\begin{aligned} A_L^{hit} &= N_U \int P(E_1) P(E_2) P_L(r) f_{D_{rU}}(r) dr \\ &= N_U \left(\int_{h_U}^{x_h} \exp(-\pi\lambda_M D_{LM}^2(r)) P_L(r) f_{D_{rU}}(r) \right. \\ &\quad \times \left[\int_r^{x_h} P_L(x) f_{D_{rU}}(x) dx \right. \\ &\quad \left. \left. + \int_{D_{LN}(r)}^{x_h} P_N(x) f_{D_{rU}}(x) dx \right]^{N_U - 1} dr \right). \end{aligned} \quad (20)$$

Thus, A_L is given by $A_L = P_{hit} A_L^{hit}$. Similar to the derivation of A_L , the association probability that the reference user is associated with an NBS, A_N , is given by $A_N = P_{hit} A_N^{hit}$, where

$$\begin{aligned} A_N^{hit} &= N_U \left(\int_{h_U}^{x_h} \exp(-\pi\lambda_M D_{NM}^2(r)) P_N(r) f_{D_{rU}}(r) \right. \\ &\quad \times \left[\int_r^{x_h} P_N(x) f_{D_{rU}}(x) dx \right. \\ &\quad \left. \left. + \int_{D_{NL}(r)}^{x_h} P_L(x) f_{D_{rU}}(x) dx \right]^{N_U - 1} dr \right). \end{aligned} \quad (21)$$

C. Conditional Serving Distance Distribution

The serving distance of a BS associated with the reference user is defined as the distance between the reference user and the associated BS. We first derive the distributions of the serving distance under different association conditions.

1) *Cache Miss Case*: According to the association strategy, the reference user is supposed to be connected to the nearest MBS under the cache miss condition, where the serving distance is equal to the distance between the reference user and the nearest MBS, denoted as R_M^{miss} . Thus, the PDF of R_M^{miss} is given by

$$f_{R_M^{miss}}(r) = 2\pi\lambda_M r e^{-\pi\lambda_M r^2}. \quad (22)$$

2) *Cache Hit Case*: According to the association strategy, the reference user can be connected to the three types of BSs in the cache hit case. Let R_L^{hit} denote the serving distance when the reference user is associated with an LBS. Thus, the conditional CDF of R_L^{hit} is given by

$$\begin{aligned} F_{R_L^{hit}|LBS}(r) &= \mathbb{P}(R_L^{hit} < r | E_{LBS}) \\ &= \frac{N_U}{A_L^{hit}} \left(\int_{h_U}^r \exp(-\pi\lambda_M D_{LM}^2(w)) P_L(w) f_{D_{rU}}(w) \right. \\ &\quad \times \left[\int_w^{x_h} P_L(x) f_{D_{rU}}(x) dx \right. \\ &\quad \left. \left. + \int_{D_{LN}(w)}^{x_h} P_N(x) f_{D_{rU}}(x) dx \right]^{N_U-1} dw \right), \end{aligned} \quad (23)$$

where E_{LBS} denotes the event that the reference user is connected to an LBS. Thus, the conditional PDF of R_L^{hit} is given by

$$\begin{aligned} f_{R_L^{hit}|LBS}(r) &= \frac{dF_{R_L^{hit}|LBS}(r)}{dr} \\ &= \frac{N_U P_L(r) f_{D_{rU}}(r) \exp(-\pi\lambda_M D_{LM}^2(r))}{A_L^{hit}} \\ &\quad \times \left[\int_r^{x_h} P_L(x) f_{D_{rU}}(x) dx \right. \\ &\quad \left. + \int_{D_{LN}(r)}^{x_h} P_N(x) f_{D_{rU}}(x) dx \right]^{N_U-1}. \end{aligned} \quad (24)$$

Let R_N^{hit} denote the serving distance when the reference user is associated with an NBS. Similar to the derivation of the conditional PDF of R_L^{hit} , the conditional PDF of R_N^{hit} is given by

$$\begin{aligned} f_{R_N^{hit}|NBS}(r) &= \frac{N_U P_N(r) f_{D_{rU}}(r) \exp(-\pi\lambda_M D_{NM}^2(r))}{A_N^{hit}} \\ &\quad \times \left[\int_r^{x_h} P_N(x) f_{D_{rU}}(x) dx \right. \\ &\quad \left. + \int_{D_{NL}(r)}^{x_h} P_L(x) f_{D_{rU}}(x) dx \right]^{N_U-1}. \end{aligned} \quad (25)$$

Let R_M^{hit} denote the serving distance when the reference user is associated with an MBS in the cache hit case. Similar to

the derivation of the conditional PDF of R_L^{hit} , the conditional PDF of R_M^{hit} is given by

$$\begin{aligned} f_{R_M^{hit}|MBS}(r) &= \frac{f_{D_{rM}}(r)}{A_M^{hit}} \times \left[\int_{D_{ML}(r)}^{x_h} P_L(x) f_{D_{rU}}(x) dx \right. \\ &\quad \left. + \int_{D_{MN}(r)}^{x_h} P_N(x) f_{D_{rU}}(x) dx \right]^{N_U}. \end{aligned} \quad (26)$$

D. Content Delivery Success Probability

The content delivery success probability of a network is defined as the probability that a reference metaverse user can successfully obtain its requested content from its associated BS, which is denoted as P_{CDS} . In the cache miss case, the reference user is supposed to be connected to the nearest MBS. Considering that an MBS can provide any content requested by the reference user, the conditional content delivery success probability is equal to the probability that the signal-to-interference-to-noise ratio (SINR) at the reference user is higher than a given threshold β when the reference user is associated with an MBS. In the cache hit case, both the MBSs and UBSs can provide the content requested by the reference user. The conditional content delivery success probability is equal to the probability that the SINR at the reference user is higher than the given threshold when the reference user is associated with an MBS/UBS.

Let P_{CDS}^{miss} denote the conditional content delivery success probability in the cache miss case. Let P_{CDS}^{hit} denote the conditional content delivery success probability in the cache hit case. Thus, the content delivery success probability of the network can be expressed as

$$P_{CDS} = (1 - P_{hit}) \times P_{CDS}^{miss} + P_{hit} \times P_{CDS}^{hit}. \quad (27)$$

Furthermore, P_{CDS}^{hit} is given by

$$P_{CDS}^{hit} = A_M^{hit} \times P_{CDS}^M + A_L^{hit} \times P_{CDS}^L + A_N^{hit} \times P_{CDS}^N, \quad (28)$$

where $P_{CDS}^M/P_{CDS}^L/P_{CDS}^N$ denotes the probability that the SINR at the reference user is higher than β when the reference user is associated with an MBS/LBS/NBS. To obtain the conditional delivery success probability, we further derive the conditional content delivery success probability P_{CDS}^{miss} , P_{CDS}^M , P_{CDS}^L , and P_{CDS}^N .

Fig. 2 illustrates the distances between the reference user, a serving MBS, and an interfering MBS, where r denotes the distance between the reference user and a serving MBS, d_{MM} denotes the distance between a serving MBS and an interfering MBS, V_{MM} denotes the distance between the reference user and an interfering MBS, φ denotes the angle between the x-axis and the vector $\vec{r}$, θ denotes the angle between the x-axis and the vector $\vec{d}_{MM}$. According to the definition of P_{CDS}^{miss} , we have

$$\begin{aligned} P_{CDS}^{miss} &= \mathbb{E}_{r,\varphi} [\mathbb{P}(\text{SINR}_M^{miss} > \beta | r, \varphi)] \\ &= \int_0^{2\pi} \int_0^\infty \frac{f_{R_M^{miss}}(r)}{2\pi} \mathbb{P}(\text{SINR}_M^{miss} > \beta | r, \varphi) dr d\varphi, \end{aligned} \quad (29)$$

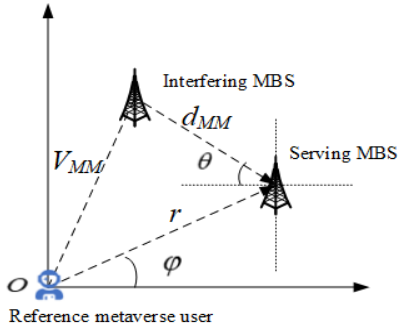


Fig. 2. Distance illustration between a reference user and MBSs.

where $SINR_M^{miss}$ denotes the SINR at the reference user from a serving MBS in the cache miss case. Moreover, the conditional content delivery success probability in Eq. (29) is given by

$$\begin{aligned}
 & \mathbb{P}(SINR_M^{miss} > \beta | r, \varphi) \\
 &= \mathbb{P}\left(\frac{P_M r^{-\alpha_M} g_M}{I_{M|M}^{miss} + I_{U|M}^{miss} + \sigma^2} > \beta | r, \varphi\right) \\
 &= \mathbb{P}(g_M > \frac{\beta r^{\alpha_M}}{P_M} (I_{M|M}^{miss} + I_{U|M}^{miss} + \sigma^2) | r, \varphi) \\
 &\stackrel{(a)}{=} \mathbb{E}_{I_{M|M}^{miss}, I_{U|M}^{miss}} \left[\exp\left(-\frac{\beta r^{\alpha_M}}{P_M} (I_{M|M}^{miss} + I_{U|M}^{miss} + \sigma^2)\right) \right] \\
 &\stackrel{(b)}{=} \mathbb{E}_{I_{M|M}^{miss}, I_{U|M}^{miss}} \left[\exp\left(-S_M (I_{M|M}^{miss} + I_{U|M}^{miss} + \sigma^2)\right) \right] \\
 &= \mathbb{E}_{I_{M|M}^{miss}} [\exp(-S_M I_{M|M}^{miss})] \\
 &\quad \times \mathbb{E}_{I_{U|M}^{miss}} [\exp(-S_M I_{U|M}^{miss})] \times \exp(-S_M \sigma^2) \\
 &\stackrel{(c)}{=} L_{I_{M|M}^{miss}}(S_M) \times L_{I_{U|M}^{miss}}(S_M) \times \exp(-S_M \sigma^2) \quad (30)
 \end{aligned}$$

where $I_{M|M}^{miss}/I_{U|M}^{miss}$ denotes the interference from interfering MBSs/UBSs when the reference user is associated with a serving MBS in the cache miss case, (a) follows from the

CCDF of g_M , (b) is obtained according to $S_M = \beta r^{\alpha_M}/P_M$, (c) follows from the definition of a Laplace transform (LT), and $\mathbb{E}_{I_{M|M}^{miss}, I_{U|M}^{miss}}[\cdot]$ denotes the expectation of $I_{M|M}^{miss}$ and $I_{U|M}^{miss}$. For the LT of $I_{M|M}^{miss}$ in Eq. (30), it is given by

$$\begin{aligned}
 & L_{I_{M|M}^{miss}}(S_M) \\
 &= \mathbb{E}_{I_{M|M}^{miss}} [\exp(-S_M I_{M|M}^{miss})] \\
 &= \mathbb{E}_{\Phi_M, g_M} [\exp(-S_M \sum_{\Phi_M} P_M V_{M_i|M}^{-\alpha_M} g_M)] \\
 &= \mathbb{E}_{\Phi_M} [\prod_{\Phi_M} \mathbb{E}_{g_M} \exp(-S_M P_M V_{M_i|M}^{-\alpha_M} g_M)] \\
 &\stackrel{(a)}{=} \mathbb{E}_{\Phi_M} [\prod_{\Phi_M} \frac{1}{1 + S_M P_M V_{M_i|M}^{-\alpha_M}}] = \mathbb{E}_{\Phi_M} [\prod_{\Phi_M} (1 - \Delta_M)] \\
 &\stackrel{(b)}{\geq} \exp(-\mathbb{E}_{\Phi_M} [\sum_{\Phi_M} \Delta_M]) \stackrel{(c)}{=} \exp[-(\Delta_1 + \Delta_2)] = \tilde{L}_{I_{M|M}^{miss}}(S_M) \quad (31)
 \end{aligned}$$

where $V_{M_i|M}$ denotes the distance between the reference user and an interfering MBS M_i when the reference user is associated with a serving MBS, $\tilde{L}_{I_{M|M}^{miss}}(S_M)$ denotes the lower bound of $L_{I_{M|M}^{miss}}(S_M)$, (a) follows from the moment generating function (MGF) of g_M , (b) is obtained according to the proposition 1 in [19], and (c) is obtained according to Campbell's Theorem in [25]. Moreover, Δ_M , Δ_1 , and Δ_2 in Eq. (31) respectively represent as in Eq. (32), Eq. (33), and Eq. (34), shown at the bottom of the page, where $\zeta_1^{(2)}(d_{MM}) = \frac{2\kappa(d_{MM})[1 - \exp(-\lambda_P \pi R_C^2)]}{\pi R_C^2 \kappa(d_{MM})[\kappa(d_{MM}) - \pi R_C^2]} - \frac{2\pi R_C^2 [1 - \exp(-\lambda_P \kappa(d_{MM}))]}{\pi R_C^2 \kappa(d_{MM})[\kappa(d_{MM}) - \pi R_C^2]}$, $\zeta_2^{(2)}(d_M) = \lambda_M^2$, $\kappa(d_{MM}) = 2\pi R_C^2 - 2R_C^2 \cos^{-1}[d_{MM}/(2R_C)] + d_{MM} \sqrt{R_C^2 - d_{MM}^2}/4$, and d_{MM} denotes the distance between the serving MBS and an interfering MBS. For the LT of $I_{U|M}^{miss}$ in Eq. (30), it is given by as in Eq. (35), shown at the bottom of the page, where (a) follows from the i.i.d. interference distances, (b) follows

$$\Delta_M = \frac{S_M P_M V_{M_i|M}^{-\alpha_M}}{1 + S_M P_M V_{M_i|M}^{-\alpha_M}} = (1 + \frac{1}{1 + S_M P_M V_{M_i|M}^{-\alpha_M}})^{-1} = \left\{ 1 + 1/[1 + S_M P_M [d_{MM}^2 + r^2 - 2rd_{MM} \cos(\theta + \varphi)]^{-\frac{\alpha_M}{2}}] \right\}^{-1} \quad (32)$$

$$\Delta_1 = \int_0^{2\pi} \int_{\max(R_C, |2r \cos(\theta + \varphi)|)}^{\max(2R_C, |2r \cos(\theta + \varphi)|)} \lambda_M^{-1} \Delta_M d_{MM} \zeta_1^{(2)}(d_{MM}) dd_{MM} d\theta \quad (33)$$

$$\Delta_2 = \int_0^{2\pi} \int_{\max(2R_C, |2r \cos(\theta + \varphi)|)}^{\infty} \lambda_M^{-1} \Delta_M d_{MM} \zeta_2^{(2)}(d_{MM}) dd_{MM} d\theta \quad (34)$$

$$\begin{aligned}
 & L_{I_{U|M}^{miss}}(S_M) = \mathbb{E}_{I_{U|M}^{miss}} [\exp(-S_M I_{U|M}^{miss})] = \mathbb{E}_{\Phi_U, g_L, g_N} \left[\exp\left(-S_M \sum_{\Phi_U} I_{U_j|M}^{miss}\right) \right] \\
 &\stackrel{(a)}{=} \left[\int_{h_U}^{x_h} [P_L(v) \mathbb{E}_{g_L} [\exp(-S_M P_U v^{-\alpha_L} g_L)] + P_N(v) \mathbb{E}_{g_N} [\exp(-S_M P_U v^{-\alpha_N} g_N)]] f_{D_{r,U}}(v) dv \right]^{N_U} \\
 &\stackrel{(b)}{=} \left[\int_{h_U}^{x_h} [P_L(v) (1 + S_M P_U v^{-\alpha_L}/m)^{-m} + P_N(v) (1 + S_M P_U v^{-\alpha_N})^{-1}] f_{D_{r,U}}(v) dv \right]^{N_U} \quad (35)
 \end{aligned}$$

from the MGF of g_L and that of g_N . For P_{CDS}^M , it is given by

$$\begin{aligned} P_{CDS}^M &= \mathbb{E}_{r,\varphi}[\mathbb{P}(\text{SINR}_M^{\text{hit}} > \beta \mid r, \varphi)] \\ &= \int_0^{2\pi} \int_0^\infty \frac{f_{R_M^{\text{hit}}|MBS}(r)}{2\pi} \mathbb{P}(\text{SINR}_M^{\text{hit}} > \beta \mid r, \varphi) dr d\varphi, \end{aligned} \quad (36)$$

where $\text{SINR}_M^{\text{hit}}$ denotes the SINR at the reference user from a serving MBS in the cache hit case. Moreover, the conditional content delivery success probability in Eq. (36) is given by

$$\begin{aligned} &\mathbb{P}(\text{SINR}_M^{\text{hit}} > \beta \mid r, \varphi) \\ &= \mathbb{P}\left(\frac{P_M r^{-\alpha_M} g_M}{I_{M|M}^{\text{hit}} + I_{U|M}^{\text{hit}} + \sigma^2} > \beta \mid r, \varphi\right) \\ &= \mathbb{P}(g_M > \frac{\beta r^{\alpha_M}}{P_M} (I_{M|M}^{\text{hit}} + I_{U|M}^{\text{hit}} + \sigma^2) \mid r, \varphi) \\ &\stackrel{(a)}{=} \mathbb{E}_{I_{M|M}^{\text{hit}}, I_{U|M}^{\text{hit}}} \left[\exp\left(-S_M (I_{M|M}^{\text{hit}} + I_{U|M}^{\text{hit}} + \sigma^2)\right) \right] \\ &= \mathbb{E}_{I_{M|M}^{\text{hit}}} [\exp(-S_M I_{M|M}^{\text{hit}})] \\ &\quad \times \mathbb{E}_{I_{U|M}^{\text{hit}}} [\exp(-S_M I_{U|M}^{\text{hit}})] \times \exp(-S_M \sigma^2) \\ &\stackrel{(b)}{=} L_{I_{M|M}^{\text{hit}}}(S_M) \times L_{I_{U|M}^{\text{hit}}}(S_M) \times \exp(-S_M \sigma^2) \end{aligned} \quad (37)$$

where $I_{M|M}^{\text{hit}}/I_{U|M}^{\text{hit}}$ denotes the interference from interfering MBSs/UBSs when the reference user is associated with a

serving MBS in the cache hit case, (a) follows from the CCDF of g_M and (b) follows from the definition of an LT. Moreover, the LT of the interference $I_{M|M}^{\text{hit}}$ from interfering MBSs in Eq. (37) is equal to $L_{I_{M|M}^{\text{hit}}}(S_M)$ in Eq. (31), and the LT of $I_{U|M}^{\text{hit}}$ in Eq. (37) is given by as in Eq. (38), shown at the bottom of the page, where (a) follows from the i.i.d. interference distances, and (b) follows from the MGF of g_L and that of g_N . For P_{CDS}^L , it is given by

$$\begin{aligned} P_{CDS}^L &= \mathbb{E}_r[\mathbb{P}(\text{SINR}_L > \beta \mid r)] \\ &= \int_{h_U}^{x_h} f_{R_L|LBS}(r) \mathbb{P}(\text{SINR}_L > \beta \mid r) dr, \end{aligned} \quad (39)$$

where SINR_L denotes the SINR at the reference user from a serving LBS in the cache hit case. Moreover, the conditional content delivery success probability in Eq. (39) is given by as in Eq. (40), shown at the bottom of the page, where $I_{M|L}^{\text{hit}}/I_{U|L}^{\text{hit}}$ denotes the interference from interfering MBSs/UBSs when the reference user is associated with a serving LBS in the cache hit case, $m_L = m(m!)^{-\frac{1}{m}}$, (a) is obtained according to $S_L = \beta r^{\alpha_L}/P_U$, (b) follows from the CCDF of g_L , (c) follows from the fact that the CCDF of a gamma random variable g_L can be approximated as a weighted sum of the CCDFs of exponential random variables [27], (d) is obtained according to $S_L^m = k m_L S_L$, and (e) follows from the definition of an LT. For the LT of $I_{M|L}^{\text{hit}}$ in Eq. (40), it is given by Eq. (41),

$$\begin{aligned} &L_{I_{U|M}^{\text{hit}}}(S_M) \\ &= \mathbb{E}_{I_{U|M}^{\text{hit}}} [\exp(-S_M I_{U|M}^{\text{hit}})] = \mathbb{E}_{\Phi_U, g_L, g_N} \left[\exp(-S_M \sum_{\Phi_U} I_{U|M}^{\text{hit}}) \right] \\ &\stackrel{(a)}{=} \left[\frac{\int_{D_{ML}(r)}^{x_h} P_L(v) \mathbb{E}_{g_L} [\exp(-S_M P_U v^{-\alpha_L} g_L)] f_{D_{rU}}(v) dv + \int_{D_{MN}(r)}^{x_h} P_N(v) \mathbb{E}_{g_N} [\exp(-S_M P_U v^{-\alpha_N} g_N)] f_{D_{rU}}(v) dv}{\int_{D_{ML}(r)}^{x_h} P_L(v) f_{D_{rU}}(v) dv + \int_{D_{MN}(r)}^{x_h} P_N(v) f_{D_{rU}}(v) dv} \right]^{N_U} \\ &\stackrel{(b)}{=} \left[\frac{\int_{D_{ML}(r)}^{x_h} P_L(v) \left(1 + \frac{S_M P_U v^{-\alpha_L}}{m}\right)^{-m} f_{D_{rU}}(v) dv + \int_{D_{MN}(r)}^{x_h} P_N(v) (1 + S_M P_U v^{-\alpha_N})^{-1} f_{D_{rU}}(v) dv}{\int_{D_{ML}(r)}^{x_h} P_L(v) f_{D_{rU}}(v) dv + \int_{D_{MN}(r)}^{x_h} P_N(v) f_{D_{rU}}(v) dv} \right]^{N_U} \end{aligned} \quad (38)$$

$$\begin{aligned} \mathbb{P}(\text{SINR}_L > \beta \mid r) &= \mathbb{P}\left(\frac{P_U r^{-\alpha_L} g_L}{I_{M|L}^{\text{hit}} + I_{U|L}^{\text{hit}} + \sigma^2} > \beta \mid r\right) = \mathbb{P}\left(g_L > \frac{\beta r^{\alpha_L}}{P_U} (I_{M|L}^{\text{hit}} + I_{U|L}^{\text{hit}} + \sigma^2) \mid r\right) \\ &\stackrel{(a)}{=} \mathbb{P}\left(g_L > S_L (I_{M|L}^{\text{hit}} + I_{U|L}^{\text{hit}} + \sigma^2) \mid r\right) \\ &\stackrel{(b)}{=} \mathbb{E}_{I_{M|L}^{\text{hit}}, I_{U|L}^{\text{hit}}} \left[\sum_{k=0}^{m-1} \frac{[m S_L (I_{M|L}^{\text{hit}} + I_{U|L}^{\text{hit}} + \sigma^2)]^k}{k!} \exp[-m S_L (I_{M|L}^{\text{hit}} + I_{U|L}^{\text{hit}} + \sigma^2)] \right] \\ &\stackrel{(c)}{\approx} \sum_{k=1}^m (-1)^k \binom{m}{k} \mathbb{E}_{I_{M|L}^{\text{hit}}, I_{U|L}^{\text{hit}}} \left[\exp[-k m_L S_L (I_{M|L}^{\text{hit}} + I_{U|L}^{\text{hit}} + \sigma^2)] \right] \\ &\stackrel{(d)}{=} \sum_{k=1}^m (-1)^k \binom{m}{k} \mathbb{E}_{I_{M|L}^{\text{hit}}} [\exp(-S_L^m I_{M|L}^{\text{hit}})] \times \mathbb{E}_{I_{U|L}^{\text{hit}}} [\exp(-S_L^m I_{U|L}^{\text{hit}})] \times \exp(-S_L^m \sigma^2) \\ &\stackrel{(e)}{=} \sum_{k=1}^m (-1)^k \binom{m}{k} [L_{I_{M|L}^{\text{hit}}}(S_L^m) \times L_{I_{U|L}^{\text{hit}}}(S_L^m) \times \exp(-S_L^m \sigma^2)] \end{aligned} \quad (40)$$

shown at the bottom of the page, where $V_{M_i|L}$ denotes the distance between the reference user and an interfering MBS M_i when the reference user is associated with a serving LBS, $\tilde{L}_{I_{M|L}^{hit}}(S_L^m)$ denotes the lower bound of $L_{I_{M|L}^{hit}}(S_L^m)$, (a) follows from the MGF of g_M , (b) is obtained according to the Jensen's inequality, and (c) is obtained according to Campbell's Theorem in [25].

For the LT of $I_{U|L}^{hit}$ in Eq. (40), it is given by Eq. (42), shown at the bottom of the page, where (a) follows from the i.i.d. interference distances, and (b) follows from the MGF of g_L and that of g_N .

For P_{CDS}^N , it is given by

$$P_{CDS}^N = \mathbb{E}_r[\mathbb{P}(\text{SINR}_N > \beta | r)] \\ = \int_{h_U}^{x_h} f_{R_N|NBS}(r) \mathbb{P}(\text{SINR}_N > \beta | r) dr, \quad (43)$$

where SINR_N denotes the SINR at the reference user from a serving NBS in the cache hit case. Moreover, the conditional content delivery success probability in Eq. (43) is given by

$$\begin{aligned} & \mathbb{P}(\text{SINR}_N > \beta | r) \\ &= \mathbb{P}\left(\frac{P_U r^{-\alpha_N} g_N}{I_{M|N}^{hit} + I_{U|N}^{hit} + \sigma^2} > \beta | r\right) \\ &= \mathbb{P}(g_N > \frac{\beta r^{\alpha_N}}{P_U} (I_{M|N}^{hit} + I_{U|N}^{hit} + \sigma^2) | r) \\ &\stackrel{(a)}{=} \mathbb{P}(g_N > S_N (I_{M|N}^{hit} + I_{U|N}^{hit} + \sigma^2) | r) \\ &\stackrel{(b)}{=} \mathbb{E}_{I_{M|N}^{hit}, I_{U|N}^{hit}} \exp[-S_N (I_{M|N}^{hit} + I_{U|N}^{hit} + \sigma^2)] \\ &\stackrel{(c)}{=} L_{I_{M|N}^{hit}}(S_N) \times L_{I_{U|N}^{hit}}(S_N) \times \exp(-S_N \sigma^2), \quad (44) \end{aligned}$$

where $I_{M|N}^{hit}/I_{U|N}^{hit}$ denotes the interference from interfering MBSs/UBSs when the reference user is associated with a serving NBS in the cache hit case, (a) is obtained according to $S_N = \beta r^{\alpha_N}/P_U$, (b) follows from the CCDF of g_N , and (c) follows from the definition of an LT.

Similarly, the LT of the interference from interfering MBSs $I_{M|N}^{hit}(S_N)$ is given by

$$\begin{aligned} & L_{I_{M|N}^{hit}}(S_N) \\ &= \mathbb{E}_{I_{M|N}^{hit}} \left[\exp(-S_N I_{M|N}^{hit}) \right] \\ &> \exp\left(-\int_{D_{NM}(r)}^{\infty} 2\pi\lambda_M \ln(1 + S_N P_M v^{-\alpha_M}) v dv\right), \\ &= \tilde{L}_{I_{M|N}^{hit}}(S_N) \quad (45) \end{aligned}$$

where $\tilde{L}_{I_{M|N}^{hit}}(S_N)$ denotes the lower bound of $L_{I_{M|N}^{hit}}(S_N)$. The LT of the interference from interfering UBSs $L_{I_{U|N}^{hit}}(S_N)$ in Eq. (44) is given by as in Eq. (46), shown at the bottom of the next page.

Let $\tilde{P}_{CDS}^{miss}/\tilde{P}_{CDS}^M/\tilde{P}_{CDS}^L/\tilde{P}_{CDS}^N$ denote the lower bound of the conditional delivery success probability $P_{CDS}^{miss}/P_{CDS}^M/P_{CDS}^L/P_{CDS}^N$. Thus, we have as in Eq. (47)–Eq. (50), shown at the bottom of the next page. Based on the above derived results, we can obtain a lower bound of the content delivery success probability, i.e., as in Eq. (51), shown at the bottom of the next page.

E. Average Content Delivery Delay

The average content delivery delay usually depends on two components: 1) the time for a serving BS to obtain the content

$$\begin{aligned} & L_{I_{M|L}^{hit}}(S_L^m) \\ &= \mathbb{E}_{I_{M|L}^{hit}} \left[\exp(-S_L^m I_{M|L}^{hit}) \right] = \mathbb{E}_{\Phi_M, g_M} \left[\exp\left(-S_L^m \sum_{\Phi_M} P_M V_{M_i|L}^{-\alpha_M} g_M\right) \right] \\ &= \mathbb{E}_{\Phi_M} \left[\prod_{\Phi_M} \mathbb{E}_{g_M} \exp\left(-S_L^m P_M V_{M_i|L}^{-\alpha_M} g_M\right) \right] \stackrel{(a)}{=} \mathbb{E}_{\Phi_M} \left[\prod_{\Phi_M} 1 / \left(1 + S_L^m P_M V_{M_i|L}^{-\alpha_M}\right) \right] \\ &= \mathbb{E}_{\Phi_M} \left[\exp\left(-\sum_{\Phi_M} \ln\left(1 + S_L^m P_M V_{M_i|L}^{-\alpha_M}\right)\right) \right] \stackrel{(b)}{\geq} \exp\left(-\mathbb{E}_{\Phi_M} \left[\sum_{\Phi_M} \ln\left(1 + S_L^m P_M V_{M_i|L}^{-\alpha_M}\right) \right]\right) \\ &\stackrel{(c)}{=} \exp\left(-\int_{D_{LM}(r)}^{\infty} 2\pi\lambda_M \ln(1 + S_L^m P_M v^{-\alpha_M}) v dv\right) = \tilde{L}_{I_{M|L}^{hit}}(S_L^m) \quad (41) \end{aligned}$$

$$\begin{aligned} & L_{I_{U|L}^{hit}}(S_L^m) \\ &= \mathbb{E}_{I_{U|L}^{hit}} \left[\exp(-S_L^m I_{U|L}^{hit}) \right] = \mathbb{E}_{\Phi_U, g_L, g_N} \left[\exp(-S_L^m \sum_{\Phi_U} I_{U|L}^{hit}) \right] \\ &\stackrel{(a)}{=} \left[\frac{\int_r^{x_h} P_L(v) \mathbb{E}_{g_L} [\exp(-S_L^m P_U v^{-\alpha_L} g_L)] f_{D_{rU}}(v) dv + \int_{D_{LN}(r)}^{x_h} P_N(v) \mathbb{E}_{g_N} [\exp(-S_L^m P_U v^{-\alpha_N} g_N)] f_{D_{rU}}(v) dv}{\int_r^{x_h} P_L(v) f_{D_{rU}}(v) dv + \int_{D_{LN}(r)}^{x_h} P_N(v) f_{D_{rU}}(v) dv} \right]^{N_U-1} \\ &\stackrel{(b)}{=} \left[\frac{\int_r^{x_h} P_L(v) \left(1 + \frac{S_L^m P_U v^{-\alpha_L}}{m}\right)^{-m} f_{D_{rU}}(v) dv + \int_{D_{LN}(r)}^{x_h} P_N(v) (1 + S_L^m P_U v^{-\alpha_N})^{-1} f_{D_{rU}}(v) dv}{\int_r^{x_h} P_L(v) f_{D_{rU}}(v) dv + \int_{D_{LN}(r)}^{x_h} P_N(v) f_{D_{rU}}(v) dv} \right]^{N_U-1} \quad (42) \end{aligned}$$

requested by a metaverse user; 2) the time for a serving BS to transmit the requested content to a metaverse user. According to the association strategy introduced in Section III-D, a reference metaverse user is only connected to a UBS in the cache hit case. An MBS can provide any requested content by backhauling to a core network through an optical fiber without any delay. Thus, the time for a serving BS to obtain the requested content is ignored here. The average content delivery delay of a reference user is defined as the time for a serving BS to transmit unit-size content to the reference user. Thus, let D_{avg} denote the average content delivery delay of a reference user. It can be expressed as

$$D_{avg} = 1/R_{avg}, \quad (52)$$

where R_{avg} denotes the average achievable rate of a user and is defined as the average maximum transmission rate that a user can obtain with unit bandwidth. According to Shannon Theory, we have

$$R_{avg} = \mathbb{E}[\log_2(1 + \text{SINR})] \quad (53)$$

where SINR denotes the signal-to-interference-to-noise ratio at the reference user. Let $R_{avg,M}^{miss}$ denote the average achievable rate given that the reference user is associated with an MBS in the cache miss case. Let $R_{avg,M}^{hit}/R_{avg,L}^{hit}/R_{avg,N}^{hit}$ denote the average achievable rate given that the reference user is associated with an MBS/LBS/NBS in the cache hit case. Thus, we have

$$R_{avg} = (1 - P_{hit}) \times R_{avg,M}^{miss} + P_{hit} \times (A_M^{hit} \times R_{avg,M}^{hit} + A_L^{hit} \times R_{avg,L}^{hit} + A_N^{hit} \times R_{avg,N}^{hit}) \quad (54)$$

To obtain the overall average achievable rate, we further derive the conditional average achievable rates $R_{avg,M}^{miss}$, $R_{avg,M}^{hit}$, $R_{avg,L}^{hit}$ and $R_{avg,N}^{hit}$.

For $R_{avg,M}^{miss}$, we have

$$\begin{aligned} R_{avg,M}^{miss} &= \mathbb{E}[\log_2(1 + \text{SINR}_M^{miss})] \\ &= \mathbb{E}[\ln(1 + \text{SINR}_M^{miss})] / \ln 2 \\ &= \left[\int_0^\infty \mathbb{P}(\ln(1 + \text{SINR}_M^{miss}) > x) dx \right] / \ln 2 \\ &= \left[\int_0^\infty \mathbb{P}(\text{SINR}_M^{miss} > e^x - 1) dx \right] / \ln 2 \\ &\stackrel{(a)}{=} \frac{1}{\ln 2} \int_0^\infty \frac{1}{c+1} \mathbb{P}(\text{SINR}_M^{miss} > c) dc \\ &= \frac{1}{\ln 2} \int_0^\infty \frac{1}{c+1} P_{CDS}^{miss}(c) dc > \frac{1}{\ln 2} \int_0^\infty \frac{1}{c+1} \tilde{P}_{CDS}^{miss}(c) dc \end{aligned} \quad (55)$$

where (a) is obtained according to $c = e^x - 1$, $\tilde{P}_{CDS}^{miss}(c)$ can be obtained by replacing β with c in Eq. (47), and $\mathbb{E}[\Omega] = \int_0^\infty P(\Omega > \omega) d\omega$ denotes the expectation of a positive random variable Ω [25]. Similarly, for $R_{avg,M}^{hit}$, $R_{avg,L}^{hit}$ and $R_{avg,N}^{hit}$, we have

$$R_{avg,M}^{hit} = \frac{1}{\ln 2} \int_0^\infty \frac{1}{c+1} P_{CDS}^M(c) dc > \frac{1}{\ln 2} \int_0^\infty \frac{1}{c+1} \tilde{P}_{CDS}^M(c) dc, \quad (56)$$

$$R_{avg,L}^{hit} = \frac{1}{\ln 2} \int_0^\infty \frac{1}{c+1} P_{CDS}^L(c) dc > \frac{1}{\ln 2} \int_0^\infty \frac{1}{c+1} \tilde{P}_{CDS}^L(c) dc, \quad (57)$$

$$R_{avg,N}^{hit} = \frac{1}{\ln 2} \int_0^\infty \frac{1}{c+1} P_{CDS}^N(c) dc > \frac{1}{\ln 2} \int_0^\infty \frac{1}{c+1} \tilde{P}_{CDS}^N(c) dc, \quad (58)$$

where $\tilde{P}_{CDS}^M(c)/\tilde{P}_{CDS}^L(c)/\tilde{P}_{CDS}^N(c)$ can be obtained by replacing β with c in Eq. (48)/Eq. (49)/Eq. (50).

Based on the above derived results, we can obtain an upper bound of the average content delivery delay, i.e.,

$$D_{avg} < \ln 2 / \tilde{R}_{avg}, \quad (59)$$

$$\begin{aligned} L_{I_{U|N}^{hit}}(S_N) &= \mathbb{E}_{I_{U|N}^{hit}} \left[\exp(-S_N I_{U|N}^{hit}) \right] \\ &= \left[\frac{\int_0^{x_h} P_L(v) \left(1 + \frac{S_N P_U v^{-\alpha_L}}{m}\right)^{-m} f_{D_{rU}}(v) dv + \int_r^{x_h} P_N(v) (1 + S_N P_U v^{-\alpha_N})^{-1} f_{D_{rU}}(v) dv}{\int_0^{x_h} P_L(v) f_{D_{rU}}(v) dv + \int_r^{x_h} P_N(v) f_{D_{rU}}(v) dv} \right]^{N_U-1} \end{aligned} \quad (46)$$

$$P_{CDS}^{miss} > \tilde{P}_{CDS}^{miss} = \int_0^{2\pi} \int_0^\infty \frac{f_{R_M^{miss}}(r)}{2\pi} \tilde{L}_{I_{M|M}^{miss}}(S_M) \times L_{I_{U|M}^{miss}}(S_M) \times \exp(-S_M \sigma^2) dr d\varphi \quad (47)$$

$$P_{CDS}^M > \tilde{P}_{CDS}^M = \int_0^{2\pi} \int_0^\infty \frac{f_{R_M^{hit}}(r)}{2\pi} \tilde{L}_{I_{M|M}^{hit}}(S_M) \times L_{I_{U|M}^{hit}}(S_M) \times \exp(-S_M \sigma^2) dr d\varphi \quad (48)$$

$$P_{CDS}^L > \tilde{P}_{CDS}^L = \int_{h_U}^{x_h} f_{R_L|LBS}(r) \sum_{k=1}^m (-1)^k \binom{m}{k} \left[\tilde{L}_{I_{M|L}^{hit}}(S_L^m) \times L_{I_{U|L}^{hit}}(S_L^m) \times \exp(-S_L^m \sigma^2) \right] dr \quad (49)$$

$$P_{CDS}^N > \tilde{P}_{CDS}^N = \int_{h_U}^{x_h} f_{R_N|NBS}(r) \tilde{L}_{I_{M|N}^{hit}}(S_N) \times L_{I_{U|N}^{hit}}(S_N) \times \exp(-S_N \sigma^2) dr \quad (50)$$

$$\begin{aligned} P_{CDS} &= (1 - P_{hit}) \times P_{CDS}^{miss} + P_{hit} \times (A_M^{hit} \times P_{CDS}^M + A_L^{hit} \times P_{CDS}^L + A_N^{hit} \times P_{CDS}^N) \\ &> (1 - P_{hit}) \times \tilde{P}_{CDS}^{miss} + P_{hit} \times (A_M^{hit} \times \tilde{P}_{CDS}^M + A_L^{hit} \times \tilde{P}_{CDS}^L + A_N^{hit} \times \tilde{P}_{CDS}^N) \end{aligned} \quad (51)$$

TABLE I
SIMULATION PARAMETERS

Parameters	Value
P_M	20W
λ_P	$10^{-5}/\text{m}^2$
P_U	5W
R_S	1000m
R_C	50m
$(\alpha_L, \alpha_N, \alpha_M)$	(2.5, 4, 4)
(a, b)	(11.95, 0.136)
m	3
N_0	$4 \times 10^{-11}\text{W}$
C_{MAX}	1000
α	0.8

where $\tilde{R}_{avg} = (1 - P_{hit}) \times \int_0^\infty \frac{1}{c+1} \tilde{P}_{CDS}^{miss}(c)dc + P_{hit} \times (A_M^{hit} \times \int_0^\infty \frac{1}{c+1} \tilde{P}_{CDS}^M(c)dc + A_L^{hit} \times \int_0^\infty \frac{1}{c+1} \tilde{P}_{CDS}^L(c)dc + A_N^{hit} \times \int_0^\infty \frac{1}{c+1} \tilde{P}_{CDS}^N(c)dc)$.

V. NUMERICAL RESULTS

In this section, we present simulation results to validate the derived analytical performance models with the proposed BS association strategy in a cache-enabled UBS-assisted cellular network. A Monte-Carlo simulation experiment was conducted using Matlab. The impacts of system parameters on the content delivery performance for metaverse users are investigated through numerical results. Moreover, we also compares the proposed BS association strategy with two benchmark strategies in terms of the average content delivery delay.

The parameter values used in the simulation experiment and the theoretical analysis are listed in Table I [20]. Fig. 3-Fig. 7 show both the performance results obtained based on the analytical models and those obtained based on the Monte-Carlo simulation experiments. It is observed that they are very close to each other, which justifies the effectiveness of the derived performance models.

A. Impacts of UBS Parameters on the Association Probability

Fig. 3 shows the corresponding association probability versus the UBS height, where $N_U = 10$, $C_U = 600$, $C_0 = 0.8C_U$. It is seen that as the UBS height increases, the association probability with an MBS decreases first and then increases, the association probability with an LBS increases first and then decreases, and the association probability with an NBS decreases. This is because with the UBS height increasing, the probability of a LoS link between the reference user and a UBS increases according to Eq. (1), which results in an increase in the association probability with an LBS and a decrease in the association probability with an NBS. The signal power received from a UBS increases as well because of a high quality LoS link between the reference user and a UBS, which results in a decrease in the association probability with an MBS. However, as the UBS height continues to increase, the distance between a UBS and the reference user would further increase to a value on which the signal power received from a UBS is lower than that from an MBS due to the increase of path loss, which results in an increase in the association probability with an MBS and a decrease in the association probability with an LBS.

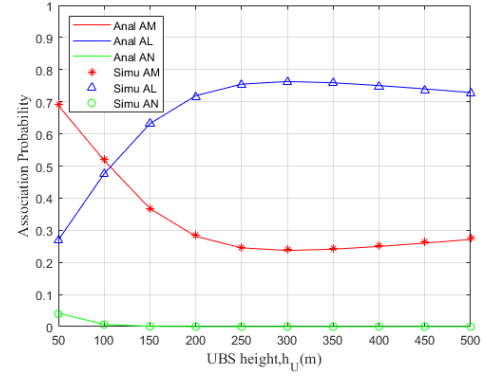
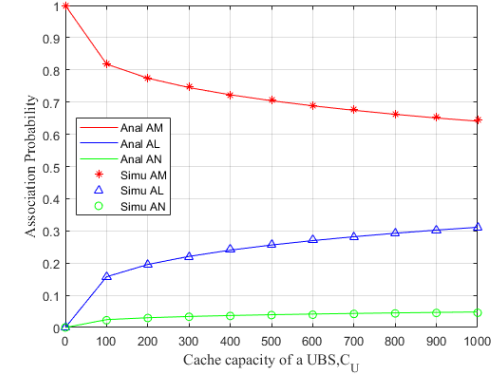
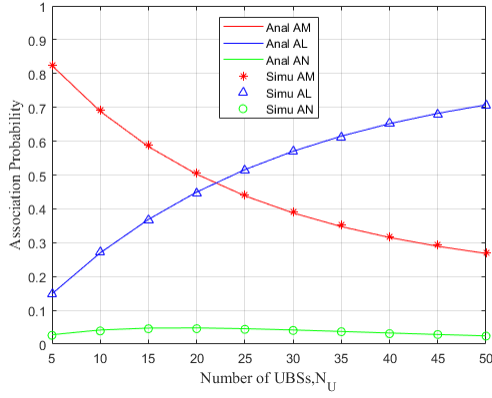
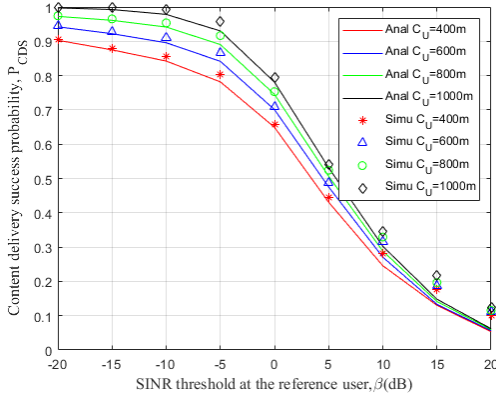
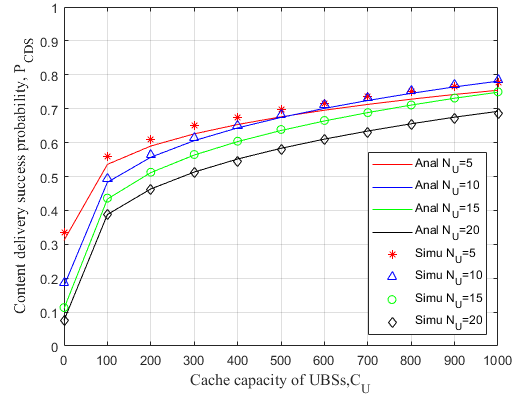
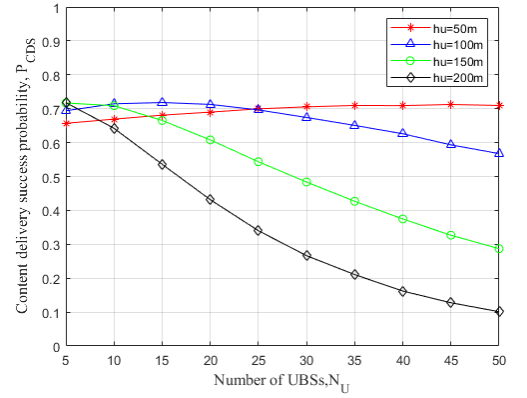

 Fig. 3. A_M, A_L, A_N vs h_U .

 Fig. 4. A_M, A_L, A_N vs C_U .

Fig. 4 shows the corresponding association probability versus the cache capacity of a UBS, where $h_U = 50\text{m}$, $N_U = 10$, $C_0 = 0.8C_U$. It is seen that as the cache capacity of a UBS increases, the association probability with an MBS decreases, the association probability with an LBS increases, and the association probability with an NBS decreases. This is because as the cache capacity of a UBS increases, the cache hit probability increases, which results in an increase in the association probability with a UBS and a decrease in the association probability with an MBS.

Fig. 5 shows the corresponding association probability versus the number of UBSs, where $h_U = 50\text{m}$, $C_U = 600$, $C_0 = 0.8C_U$. It is seen that as the number of UBSs increases, the association probability with an MBS decreases, the association probability with an LBS increases, and the association probability with an NBS increases first and then decreases. This is because as the number of UBSs increases, a reference user has a higher probability to receive a stronger signal from a UBS than an MBS, which results in a decrease in the association probability with an MBS. Moreover, as the number of UBSs continues to increase, the LoS probability of the link between a reference user and a UBS increases, which results in a decrease in the association probability with an NBS and an increase in the association probability with an LBS.

B. Impacts of System Parameters on the Content Delivery Success Probability

1) *SINR Threshold*: Fig. 6 shows the content delivery success probability vs SINR threshold for different values of

Fig. 5. A_M, A_L, A_N vs N_U .Fig. 6. P_{CDS} vs β .Fig. 7. P_{CDS} vs C_U .Fig. 8. P_{CDS} vs N_U .

the cache capacity of a UBS, where $h_U = 150m$, $N_U = 10$, and $C_0 = 0.8C_U$. It is observed that with the SINR threshold increasing the content delivery success probability decreases. Moreover, a higher value of the cache capacity of a UBS would result in a large value of the content delivery success probability.

2) *UBS Parameters*: Fig. 7 shows the content delivery success probability versus the cache capacity of a UBS for different values of the number of UBSs, where $\beta = 0dB$, $h_U = 150m$, $C_0 = 0.8C_U$. It is observed that with the cache capacity of a UBS increasing, the content delivery success probability increases. This is because the cache hit probability increases with the cache capacity of a UBS increasing, which results in the increase in the association probability with an LBS according to Fig. 4. Accordingly, the SINR at the reference user increases with the signal power received from a serving UBS increasing.

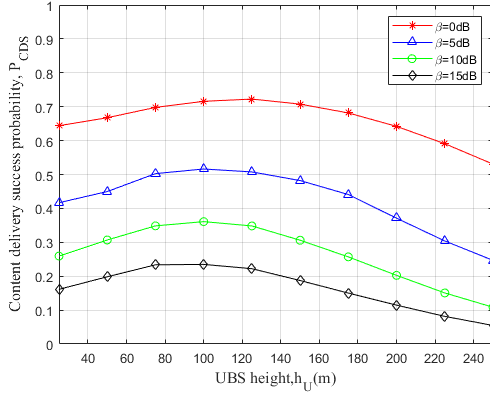
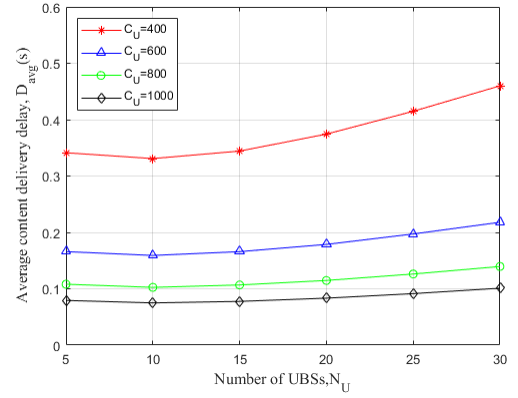
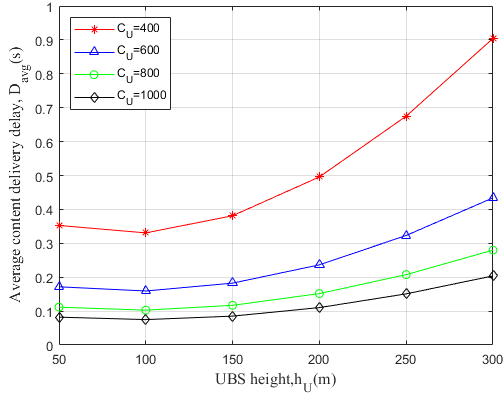
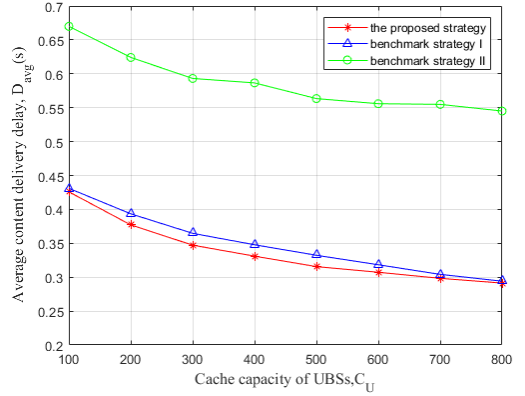
Fig. 8 shows the content delivery success probability versus the number of UBSs for different values of the UBS height, where $\beta = 0dB$, $C_U = 600$, $C_0 = 0.8C_U$. It is observed that with the number of UBSs increasing, the content delivery success probability increases first and then decreases. This is because the probability that the reference user is associated with an LBS increases with the number of UBSs increasing, which results in an increase in the signal power received from a serving UBS. However, as the number of UBSs continues to increase, the number of interfering UBSs increases and

the SINR at the reference user decreases, which results in a decrease in the content delivery success probability.

Fig. 9 shows the content delivery success probability versus the UBS height for different values of the SINR threshold, where $N_U = 10$, $C_U = 600$, $C_0 = 0.8C_U$. It is observed that with the UBS height increasing, the content delivery success probability increases first and then decreases. The reason is that with the UBS height increasing, the probability that the reference user is associated with an LBS increases as well, which results in an increase in the signal power received from a serving UBS. However, as the UBS height continues to increase, the distance between a UBS and the reference user would further increase to a value on which the SINR at the reference user starts to decrease due to the increase of path loss. Beyond the corresponding UBS height, the power gain of a signal brought by a LoS link between a UBS and the reference user would no longer increase, and thus the signal power received from a serving UBS starts to decrease. Accordingly, there exists an optimal UBS height which results in the largest content delivery success probability of the reference user.

C. Impacts of UBS Parameters on the Average Content Delivery Delay

Fig. 10 shows the average content delivery delay versus the UBS height for different values of the cache capacity of a

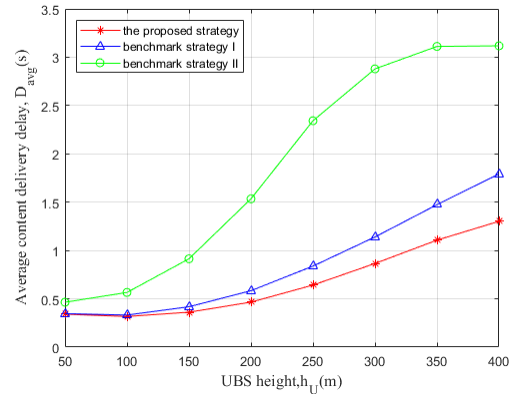

 Fig. 9. P_{CDS} vs h_U .

 Fig. 11. D_{avg} vs N_U .

 Fig. 10. D_{avg} vs h_U .

 Fig. 12. D_{avg} vs C_U .

UBS, where $C_0 = 0.8C_U$ and $N_U = 10$. It is observed that the average content delivery delay decreases first and then increases with the UBS height increasing. This is because with the UBS height increasing, the SINR at the reference user increases first due to the increase in the probability that the reference user is associated with an LBS, which results in the increase in the average achievable rate, and decreases due to the increase in the path loss, which results in the decrease in the average achievable rate.

Fig. 11 shows the average content delivery delay versus the number of UBSs for different values of the cache capacity of a UBS, where $C_0 = 0.8C_U$ and $h_U = 100m$. It is observed that the average content delivery delay decreases first and then increases with the number of UBSs increasing. This is because, with the number of UBSs increasing, the SINR at the reference user increases first due to the increase in the signal power received from a serving LBS, which results in the increase in the average achievable rate, and then decreases due to the increase in the aggregate interference from interfering UBSs, which results in the decrease in the average achievable rate.

D. Comparison of Different BS Association Strategies

Fig. 12-Fig. 14 shows the average content delivery delay versus different system parameters with the proposed strategy, benchmark strategy I, and benchmark strategy II. In benchmark strategy I, the reference user is associated with


 Fig. 13. D_{avg} vs h_U .

the BS which provides the strongest average received power in both the cache hit and cache miss case. In benchmark strategy II, the reference user is associated with the nearest BS in the cache hit case and with the nearest MBS in the cache miss case.

It is seen that the average content delivery delay with the proposed strategy is smaller than those with both benchmark strategy I and benchmark strategy II. This means that a metaverse user can obtain better content delivery delay performance using the proposed BS association strategy.

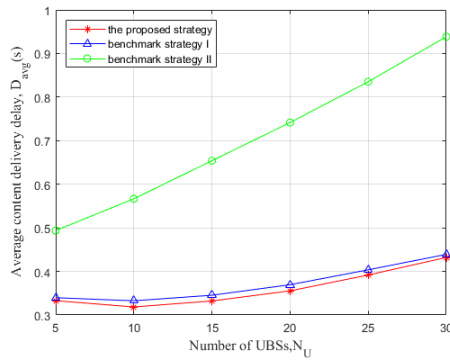


Fig. 14. D_{avg} vs N_U .

VI. CONCLUSION

In this paper, we studied the content delivery performance analysis of a cache-enabled UBS-assisted cellular network for metaverse users. Two analytical models were derived for investigating the content delivery performance of the network in terms of the content delivery success probability of the network and the average content delivery delay of a user. In deriving the analytical models, both a LoS link and an NLoS link were considered. Moreover, an MHCPP was used to model the location distribution of MBSSs. The cache hit probability of a UBS based on the caching strategy was also considered. A BS association strategy for delay-sensitive users based on the strongest average received power and the cache hit probability was proposed. In addition, the association probabilities with the association strategy were derived for different types of base stations. Finally, a lower bound of the content delivery success probability and an upper bound of the average content delivery delay were obtained from the derived analytical models. The numerical results justify the effectiveness and advantage of the proposed BS association strategy and show that there exist an optimal UBS height and an optimal value of the number of UBSs, which result in the optimal content delivery performance. The obtained results can provide theoretical guidance for the deployment of UBSs in a cache-enabled UBS-assisted cellular network to provide better human-centric content delivery service for metaverse users.

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